Smart Parking System using IoT

Hackathon Embedded SystemsProject

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Problem Statement

Findingparkinginbusyareasremains a persistentchallenge, creating inefficiencies that impact both individuals and cities.

Time Wastage

Drivers spend excessive time circling busy areas searching for empty slots

Traffic Congestion

Inefficient parking contributes significantly to urban traffic jams and pollution

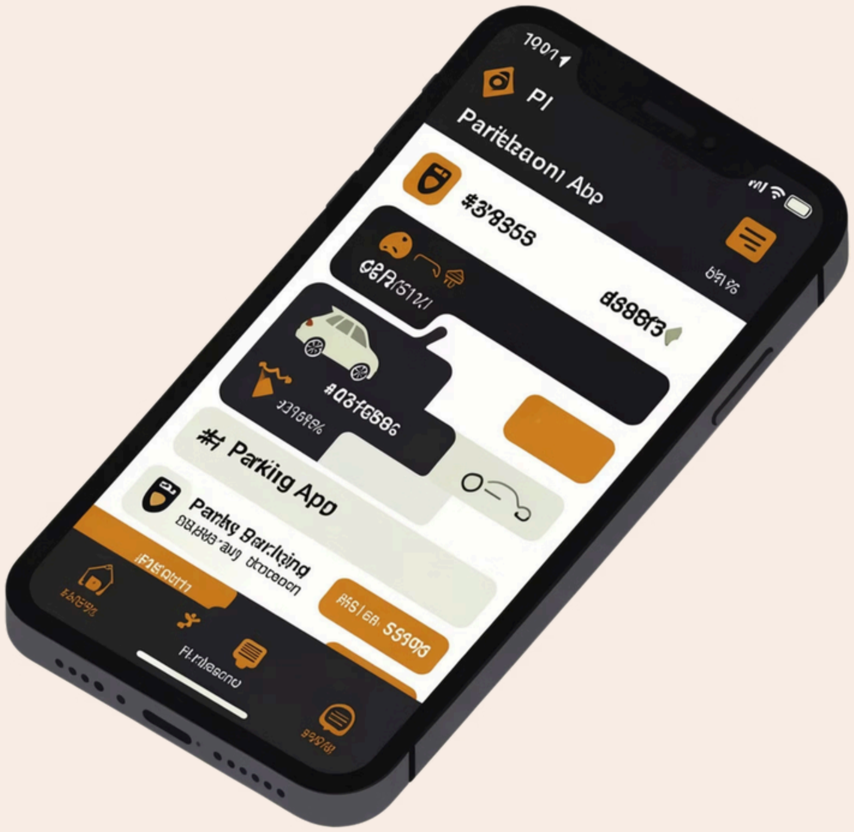
Information Gap

Lack of real-time visibility into parking availability frustrates users

Manual Processes

Existing systems require manual intervention and lack automation

Existing Solutions & Limitations



Current Approach

Appslike Park+ provide basic parking services including booking and general availability information. However, these solutions fall short of true smart parking.

Key Limitations

→ **No Slot-Level Detection**

General area availability without specific slot identification

→ **Limited Real-Time Accuracy**

Delayed updates create confusion and missed opportunities

→ **No Automated Control**

Manual gate operation and entry management required

→ **No Guidance System**

Users must search for nearest available slot themselves

Proposed Solution

Real-Time Detection

Sensorsprovideinstantslotavailability updates

Automatic Gate Control

Servomotorenablesautomatedentry based on availability

Web Application

Mobile-friendlyinterface displays parking status clearly

Smart Indication

Visualindicatorsshowfull or available status instantly

Goal: Reduce search time, traffic congestion, and manual effort through intelligent automation

System Architecture

OurIoT-basedsystemfollows a streamlineddataflowfromphysical detection to user interface.

Detect Vehicles

Process Signals

Determine Availability

Communicate & Display

Implementation Details

Hardware Components

- Raspberry Pi Pico / Pico W (microcontroller)
- Ultrasonic Sensors (3 slots monitored)
- IR Sensor (entry point detection)
- Servo Motor (gate control mechanism)
- 7-Segment Display (status indication)

Software Stack

- MicroPython programming language
- Distance-based slot detection algorithm
- Web application using Windsurf framework
- HTML, CSS, JavaScript for frontend

Core Logic

1

Distance > 10 cm

Slot marked as Empty

2

Distance < 10 cm

Slot marked as Occupied

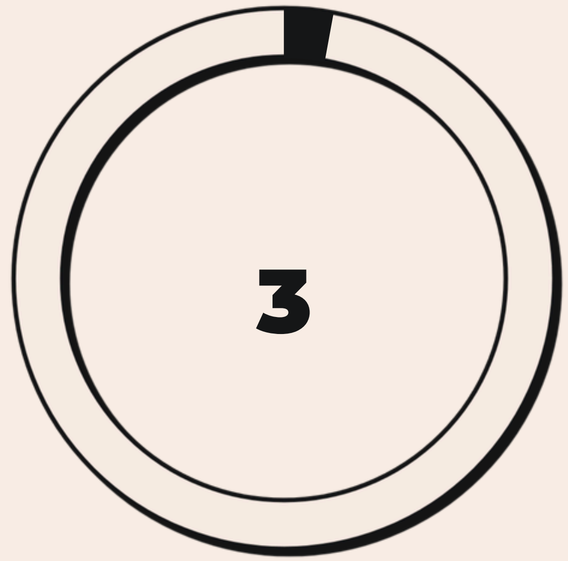
3

Availability Check

Gate opens only if slots available

Results & Demo

Real-Time Detection
Instant slot detection working reliably



Slots Monitored
Multiple parking positions tracked

Key Features Demonstrated

- Real-time slot detection successfully implemented
- Available slots displayed on 7-segment display
- Web app shows slot status with clear indicators
- Green = Empty slot
- Red = Occupied slot
- Live monitoring with instant updates

Find Nearest Slot: System automatically guides users to closest available parking position

Conclusion

Efficient Mana gem ent

Intelligent system optimizes parking space utilization and reduces search time significantly

Traffic Reduction

Minimizes congestion caused by vehicles searching for parking in busy areas

Cost-Effective

Usesaffordable, readily available components making it accessible for widespread adoption

Our IoT-based solution addresses critical parking challenges through automation, real-time monitoring, and intelligent slot allocation. The system successfully demonstrates feasibility for deployment in urban parking facilities.